

(Nathan) Khoi Nguyen Trinh

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EDUCATION

University of Illinois Chicago Expected Graduation: **May 2026**
Bachelor of Science in Computer Science, Minor in Business Administration GPA: **4.0**
Relevant Coursework: Advanced Data Structures, Algorithms, Software Design, Secure Web App Applications, Artificial Intelligence, Machine Learning, Natural Language Processing, Android Development (CodePath)

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, Kotlin, Dart, PHP
Frameworks/Platforms: React, Next, Node, Vite, Astro, Tailwind, Yii, Spring, Bootstrap, Flutter, Flask, Jest, Maven
Technologies: HTML/CSS, Git, ASAN, RESTful APIs, JWT, OAuth, MariaDB, SQLite, PostgreSQL, Firebase, Postman

EXPERIENCE

Web Development Team Lead | SparkHacks July 2025 - Present

- Lead design and full-stack development of SparkHacks 2026's main website, providing event information and registration access for **400+** participants, sponsors, mentors, and organizers.
- Enhance an admin dashboard backed by Firestore to aggregate, search, and filter **500+** records, with live analytics displaying acceptance rates, demographic breakdowns, and check-in progress.
- Manage Agile sprints and peer programming reviews across a 6-member development team, managing version control, code quality, and consistency through CI workflows.

Undergraduate Research Assistant | UIC ELiCIT Lab May 2025 - Present

- Design **60+** low-fidelity wireframes screens to lead development of an interactive AI literacy educational app, teaching children aged 8-12 privacy concepts through puzzle-solving mechanics.
- Collaborate with a PhD student and an Assistant Professor to align technical implementation with research objectives.

Data Structures Teaching Assistant | UIC Department of Computer Science January 2025 - Present

- Mentor **400+** students in C++ data structures and algorithmic problem-solving, offering course material and project support during lecture and weekly office hours.
- Assist in weekly lab sessions for **50+** students, covering complex concepts such as binary trees, hash tables, graph algorithms, and searching & sorting techniques.
- Guide students in using tools such as LLDB, ASAN, and GoogleTest to improve unit testing and debugging skills.

Software Developer Intern | Viettel Digital June 2024 - August 2024

- Developed RESTful APIs for user interaction in PHP, Spring and Yii for an integrated Fintech social network system, increasing engagement and connectivity for over **24 million** users.
- Implemented JWT and OAuth 2.0 for secure, role-based API authentication and authorization across users, merchants, and admins, enhancing security and system performance.
- Improved database performance by **40%** through optimized query optimization and API validation with Postman.

Software Developer Intern | VISSOFT July 2023 - August 2023

- Adapted the Bloom Filter algorithm for fast data retrieval, reducing lookup time and cutting memory usage by **80%** in a Java-based library management system with a MySQL backend.
- Integrated the GoogleTest framework to develop **20+** unit tests, ensuring functionality through automated testing.

PROJECTS

SparkHacks 2026 Website & Dashboard | Astro, TypeScript, Firebase | [Website](#) / [Dashboard](#) July 2025 - Present

- Build and lead full-stack development for a responsive, component-driven architecture website using Astro, reducing load time by **40%** compared to traditional React SPAs.
- Update the Firestore-backed dashboard with a dynamic QR code system to perform check-in for 400+ applicants, reducing lookup time to **2 seconds** per attendee.
- Develop an automated waitlist promotion system that monitors acceptance counts and upgrades waitlisted applicants when spots open, maintaining **98%+** attendance accuracy.

Restockd | Python, React, Flask, PostgreSQL | [Website](#) October 2025 - Present

- Develop a full-stack food bank donation platform that enables real-time matching between food banks and donors, reducing coordination time by **60%** compared to manual methods.
- Implemented advanced search optimization using Trie data structures and Bloom filters, enabling food banks to search 100+ active donation postings with **sub-50ms** query response times.
- Architected a dual-role authentication system with Supabase OAuth and role-based access control, managing meetup scheduling and time-change requests with **20+** RESTful API endpoints.

TariffFarm (WildHacks 2025 Winner) | Python, Flask, Next.js, Gemini | [GitHub](#) April 2025

- Collaborated in a team of 4 to develop a web app that calculates and visualizes U.S. import costs for agricultural commodities, winning **3rd place** overall at WildHacks 2025.
- Engineered a Flask backend with API endpoints and data pipelines to process real-time trade data, and adapted cost calculation algorithms to suggest shipping routes with Gemini.
- Developed an interactive frontend using **Next.js** and **amCharts**, delivering real-time 3D globe visualizations and a dynamic, responsive UI for an intuitive experience.